AI-assisted multi-agent quantitative fund

FE5209 project scope and design

Three strategy PMs managing a USD 5m paper fund

Proposed design
14 October 2026

Project scope

A portfolio experiment with a visible AI review workflow

Course requirements

- Automated model updates and daily forecasts
- Mandatory class assets with eligible history
- Out-of-sample evaluation: 1–23 October 2026
- Code, closing-price CSV and presentation

Proposed first version

- Three PMs with separate paper ledgers
- Shared allocation and fund risk controls
- AI agents inspect tools and cite evidence
- One hosted application with saved run history

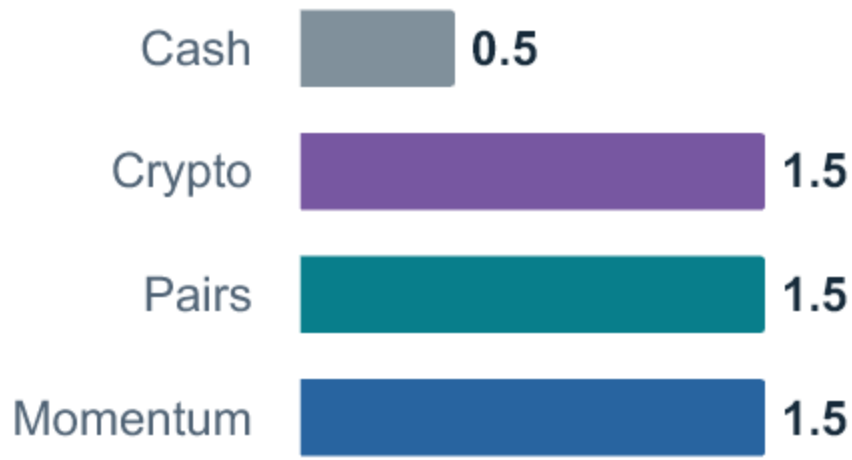
Initial capital and execution assumptions need instructor confirmation

USD 5m fund allocation

Illustrative initial mandates, with USD 0.5m held centrally

Capital owner	USD	Share
Momentum PM	1,500,000	30%
Pairs PM	1,500,000	30%
Crypto PM	1,500,000	30%
Central cash	500,000	10%
Total fund	5,000,000	100%

Initial capital, USD millions



The baseline keeps capital in each sleeve. Returns cause fund weights to drift.

Equal initial capital can produce different risk. Each PM can retain cash.

Each PM manages its own portfolio

A capital mandate gives each strategy an accountable virtual account

PM	Strategy responsibility	Cadence
Equity momentum	Run the teammate’s stock-selection rule and maintain target equity holdings	To confirm
Pairs trading	Run the teammate’s paired-position rule and preserve the relationship between legs	Daily confirmed
Crypto	Run the selected crypto strategy and manage exposure or cash	Daily proposed

Every PM tracks cash, positions, model state and net performance

Common proposal: decision time, input cutoff, model version, target holdings and reason codes

The multi-AI contribution

Each PM combines its existing strategy with a planned tool-using AI representative

AI representative	Diagnostic tools	Evidence-linked output
Momentum PM AI	Read rankings, holdings and sector exposure	Explain selections and concentration concerns
Pairs PM AI	Read daily spread signals, hedge ratios and pair health	Explain both legs and relationship concerns
Crypto PM AI	Read signals, volatility, liquidity and trading costs	Explain exposure or cash and data concerns

AI chooses investigations and writes PM memos

Quantitative code calculates targets. The coordinator and risk controller govern execution.

An AI investment committee in practice

Illustrative exchange to replace with actual agent traces in the demonstration

AI representative	Chosen diagnostic	Example PM memo
Momentum PM AI	Inspect sector exposure	Several proposed holdings share a sector. I flag concentration for fund review.
Pairs PM AI	Inspect both target legs	The target legs retain the hedge ratio. I attach the pair-health diagnostics.
Crypto PM AI	Read entry qualification	The strategy has no qualified entry. Its quantitative target stays in cash.

The coordinator combines targets. The risk controller records its verdict.

Each factual claim links to a saved record. An optional reviewer checks the PM memos.

Daily pairs decision cycle

Pairs frequency is confirmed. Exact signal and fill timing remains to be agreed.

When	Owner	Output
Session t, after close	Data adapter	Completed prices and validated timestamps
Session t, after close	Pairs quant engine	Daily model update and linked target legs
Before execution	Coordinator and risk	Combined exposure and approval verdict
Next eligible session	Paper executor	Simulated fills under the declared cost model
After valuation	Accounting and AI PMs	Fund NAV, CSV and evidence-linked memos

Both legs retain their hedge relationship. A daily signal can lead to no trade.

Capital, risk and accounting

The coordinator proposes budgets. The risk controller has final approval.

Portfolio coordinator

- Assign the initial PM capital mandates
- Combine holdings and identify overlap
- Track PM contribution to fund P&L
- Log transfers separately from returns

Independent risk controller

- Check concentration and gross/net exposure
- Preserve linked pair positions
- Check available cash, margin and data
- Reject breaches and record the reason

Fund NAV = sum of PM NAVs + central cash

Capital transfers preserve fund NAV. Costs enter the ledger once.

Performance evidence

Every consolidated result reconciles to the PM ledgers

Required course KPIs

- Net P&L, total return and annualized return
- Sharpe ratio and annualized volatility
- Maximum drawdown, its frequency and duration
- Win rate, profit per trade, trade count and holding period

Fund and PM reporting

- Fund NAV against the official benchmark
- Each PM's contribution to profit and drawdown
- Fees, slippage and applicable financing costs
- Capital transfers recorded separately from returns

Saved evidence: closing prices, signals, orders, fills, allocation decisions and AI tool traces

Evaluation plan

Trading performance and AI review quality have separate tests

Experiment	Question	Main evidence
Each PM alone	Profitable after costs?	Net return, Sharpe and drawdown
Initial-budget blend	Better as a combined fund?	Fund NAV, correlation and PM P&L
Allocation alternative	Better after reallocation costs?	Incremental return, risk and turnover
AI enabled / disabled	Accurate and useful explanations?	Cited facts, review time and cost

The AI comparison keeps quantitative portfolio decisions identical

Use the official benchmark and consistent dates, costs and capital conventions.

Delivery and ownership

One application hosts the PM views and the daily decision history

Submission and demonstration

- Reproducible strategy code and closing-price CSV
- Fund and PM performance dashboard
- Actual tool calls, risk verdicts and PM memos
- An isolated replay of a selected historical day

Proposed hosting

- Streamlit dashboard
- One Python engine with a scheduled runner
- SQLite on persistent storage
- Server-side data and model credentials

Ownership: three PM owners, an integration lead and a reporting/operations lead

For four members, share reporting and operations. Final slides must name every member.

Readiness and remaining inputs

The combined fund and connected AI PMs are proposed work

Existing foundation

- Crypto data and daily paper workflow
- Agent orchestration, risk checks and logs
- Recorded intents and optional AI replies

Inputs before integration

- Strategy packages, assets and momentum cadence
- Mandatory assets and official benchmark
- Capital, costs, shorting and execution rules
- Member names and individual responsibilities

First version: initial budgets, shared accounting, fund risk and AI tool access

Extensions: dynamic allocation, regime routing and physical order netting